

Fuzzy Information Quantity Measurement and Feature Selection by Macrogranular Entropy

Zhilin Zhu, Chucai Zhang, Jianhua Dai*

Abstract— Feature selection is an important data preprocessing process in artificial intelligence, which aims to eliminate redundant features while retaining essential features. Measuring feature significance and relevance between features is a significant challenge. Fuzzy information entropy is an extension of Shannon entropy. It is widely used for quantifying the information of fuzzy divisions. However, it has significant limitations, notably the lack of monotonicity in fuzzy conditional entropy measure of decision uncertainty in the feature selection process. We introduce a novel measurement Macrogranular Entropy (ME) and construct some generalized forms, such as conditional macrogranular entropy, mutual macrogranular information, and joint macrogranular entropy. The conditional macrogranular entropy exhibits monotonicity when measuring decision uncertainty. Additionally, we propose two feature selection algorithms: one based on Monotonic Conditional Macrogranular Entropy (MCME), and the other based on the Degree of Symmetric Association (ADSA). The ADSA algorithm and the MCME algorithm are compared against eight other feature selection algorithms through a series of experiments. The comparison was conducted based on classification performance using SVM and NB classifiers, as well as evaluation metrics including F1-score and Recall. In terms of all four evaluation metrics, ADSA and MCME achieved the top two rankings, respectively. Specifically, on the NB and SVM classifiers, the ADSA algorithm improves the average accuracy by 12.22% and 2.88% compared to the original feature set, while MCME improves the accuracy by 10.07% and 1.01%, respectively. Experimental comparisons demonstrate that ADSA algorithm effectively removes redundant information from the dataset during feature selection.

Impact Statement—We construct a novel fuzzy information measurement called Macrogranular Entropy. Macrogranular entropy addresses the theoretical limitations in fuzzy information entropy, such as the non-monotonicity of fuzzy conditional entropy. Macrogranular entropy measurement provides a new perspective by considering the union of sample relationships. In the experimental section, we use conditional macrogranular entropy to evaluate the significance of feature, with a monotonic measure of the uncertainty change in the decision. We introduce the Degree of Symmetric Association (DSA), which not only considers the proportion of the mutual information increment between the currently evaluated feature and the decision, but also takes into account the redundancy with other features. Comparative experiments between the ADSA and MCME algorithms and eight different feature selection algorithms demonstrate the

effectiveness of Macrogranular Entropy (ME) and the Degree of Symmetric Association (DSA).

Index Terms—Information measure; fuzzy information entropy; fuzzy set; feature selection; macrogranular entropy.

I. INTRODUCTION

WITH the development of the information age, data becomes increasingly complex, leading to the presence of redundant and noisy information within data. The complexity of the data can cause model training to deviate from underlying patterns, leading to poor performance [1], [2]. Redundant features information can undermine the model's learning capacity and reduce its generalization ability [3], [4], [5]. If these challenges are not addressed during the feature selection process, the model may not only perform poorly but also become prone to errors in practical applications [6], [7]. Therefore, recognizing and reducing the impacts of information redundancy is crucial for enhancing the reliability and robustness of models [3], [8], [9].

Humans often quantify information through the concept of information granularity [10], [11]. Many researchers utilize granulation techniques to measure information [12], [13], [14], [15], while also applying theories such as information entropy and rough set theory to evaluate the underlying relationships between feature information, including implication, redundancy, and interaction [4], [14], [7], [16]. High-dimensional data frequently comprises a multitude of features that exhibit intricate interrelations, thereby giving rise to a complex feature space. This inherent complexity increases the computational burden on feature selection algorithms, making it challenging to identify informative features efficiently [17], [18], [19]. Redundant features exacerbate the problem by introducing noise and decreasing the overall efficiency of model [20], [12]. Information redundancy can lead to overfitting, where the model learns noise instead of underlying patterns, thus reducing its generalization ability [21], [22], [23], [24]. The presence of redundant features can result in a significant amount of storage space being consumed, leading to wasted storage resources [11], [25], [26].

Granular computing [7], [16], [17], [27], by granulating data information, decomposes large-scale problems into smaller subproblems, aligning with human cognitive processes for effective analysis and problem-solving in real-world scenarios. This approach enhances the capability to address complex, large-scale challenges [27], [28], [29]. In recent years, researchers have developed various models that consider different levels of granularity, aiming to preserve essential

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information in the data while simultaneously reducing its dimensionality [2], [30], [31].

The Pawlak rough set model [32], as an important granular computing model, partitions sample set into several classes through the equivalence relation. It is effective to handle symbolic data tasks, but not perform well in real-valued data tasks. Dubois et al. [33] introduced the fuzzy rough set model, which constructs similarity relations between samples, eliminating the need for data discretization and enabling effective application to real-valued tasks. Qian et al. [27] proposed the concept of fuzzy information granularity in fuzzy binary models. Wang et al. [34] also proposed a fuzzy fitting model, which fits the fuzzy similar classes with the decision classes. Huang et al. [17] proposed a multi-scale fuzzy decision system that formally represents knowledge at different scales for handling of real-world data by selecting appropriate fuzzy granularity. Chen and Lin et al. [21] based kernel fuzzy rough set model and spectral graph theory, utilizing kernel fuzzy approximation operators to establish within-class and between-class sample similarity relations. Gao et al. [22] introduced a novel framework by considering the impact of granule size, combining stability theory from machine learning with the fuzzy rough set model. Wan et al. [24] proposed a robust fuzzy rough approximation space based on graph theory research combined with fuzzy rough set theory. Dai et al. [16] proposed a fuzzy implication granularity information measurement based on fuzzy rough set model.

Measuring the quantity of information in real-valued data is a significant issue [5], [8], [13], [35]. A primary approach involves extending Shannon entropy [36], as demonstrated by Yager et al. [37], to enable its application in quantifying information within real-valued features. Hu et al. [38] further extended Yager entropy and defined various forms to construct fuzzy information entropy measurements, which can effectively quantify the information quantity in real-valued tasks [4], [5], [14], [35]. Dai et al. [39] propose a multi-label feature selection optimization framework based on fuzzy mutual information incorporate global correlation for enhanced efficiency and accuracy. Wang et al. [28] proposes a multiscale fuzzy entropy-based feature selection algorithm that integrates granule information across different scales. Ju et al. [40] propose a feature selection algorithm based on a dual-channel fuzzy interaction information fusion, combining sparse granularity and shared granularity. Wang et al. [41] introduce a feature selection algorithm, MRMSR, which improves classification accuracy by maximizing feature relevance and minimizing supervisory redundancy.

In feature selection process, conditional entropy and mutual information are commonly used to measure the importance of features for decision making [2], [4], [42], [29]. A notable limitation of fuzzy conditional entropy and fuzzy mutual information [38] is their lack of monotonicity in measuring decision uncertainty as features are incrementally selected. Consequently, when dealing with real-valued features, this lack of monotonicity can make the termination of the feature selection process uncontrollable, as well as affecting the reasonableness and interpretability of the selected subset of features. Dai et al. [6] identified the absence of monotonicity as

a critical limitation, emphasizing that it undermines the theoretical foundation for establishing rational stopping conditions in the feature selection process.

Many researchers use the magnitude of mutual information between features and the decision as a approach of feature significance, evaluating the relevance between features and the decision [3], [7], [39]. However, solely using mutual information to measure feature significance fails to distinguish between two features with the same quantity of mutual information. Witten et al. [43] propose using symmetric uncertainty to measure feature significance by considering the proportion of mutual information relative to the total information quantity. Zeng et al. [8] further extend this approach and achieve promising results in feature selection tasks. However, these measurements lack consideration of the redundancy between features and candidate features, focusing solely on the relevance between features and the decision. Information redundancy between features can reduce the generalization ability of the final feature subset [7], [39]. Therefore, there are still urgent issues in real-valued feature selection tasks that require further improvement.

Inspired by these issues, we propose a novel information measurement Macrogranular Entropy (ME) and an innovative feature significance metric Degree of Symmetric Association (DSA). The main contributions of this paper be summarized as follows:

(1) We first introduce the basic concepts and generalised forms of fuzzy information entropy measurement, highlighting the drawback of the lack of monotonicity of fuzzy conditional entropy in measuring decision uncertainty. We propose a new information measure, Macrogranular Entropy (ME). Conditional macrogranular entropy possesses monotonicity, and we provide a more intuitive comparison with fuzzy information entropy.

(2) We present a series of Macrogranular Entropy forms to evaluate relationships between features, including conditional macrogranular entropy, mutual macrogranular information, and joint macrogranular entropy. Additionally, we define the relationships among these measurements and their associated propositions. To demonstrate the superiority of macrogranular entropy, we provide relevant examples and supporting evidence.

(3) We propose two feature selection algorithms based on macrogranular entropy: one is based on the monotonic conditional macrogranular entropy (MCME), and the other is based on the degree of symmetric association (ADSA) to address the deficiencies of mutual information in evaluating the relationship between features and the decision.

(4) We conduct experiments on 18 public datasets, comparing our algorithms with eight other feature selection algorithms. Additionally, we perform an in-depth analysis of the experimental results to demonstrate the advantages of the proposed macrogranular entropy measure and the degree of symmetric association (DSA) in evaluating feature significance.

The structure of the paper is organized as follows: In section II, we briefly summarize the fuzzy information entropy. In section III, we propose a novel measurement macrogranular

entropy, along with its generalized forms and propositions. Furthermore, we propose the Degree of Symmetric Association (DSA) to comprehensively measure feature significance, and we construct the feature selection algorithms based on DSA and conditional macrogranular entropy. In Section IV, we evaluate the superiority of our algorithms through extensive experiment and statistical analysis. In the conclusion V, we summarize the main contributions of this paper and further highlight the potential future work.

II. PRELIMINARY KNOWLEDGE

In this section, we primarily introduce the foundational knowledge of fuzzy information entropy measurement proposed by Hu et al. [38], as well as the concept of symmetric uncertainty proposed by Witten et al. [43] to address the limitations of mutual information.

A. Fuzzy Information Measurement

This subsection primarily introduces the fuzzy information entropy [38] for measuring information of features, along with its generalized forms.

For a given fuzzy decision table $FDT = \langle U, C \cup D, V \rangle$, where U is a finite, non-empty sample set, and C is the feature set, for a given feature subset $B \subseteq C$. Based on the concept of fuzzy rough sets, we can construct a relationship between samples in the feature subset $B \subseteq C$, represented by a matrix $M(\mathcal{R}_B)$ of size $n \times n$, where each matrix element r_{ij} represents the similarity of two samples x_i, x_j .

Definition 1. [38] Given an $FDT = \langle U, C \cup D, V \rangle$, for a feature subset $B \subseteq C$, then the fuzzy information entropy of B is defined as:

$$H(\tilde{B}) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|\tilde{[x_i]_B}|}{n}, \quad (1)$$

where $|\tilde{[x_i]_B}|$ is the cardinality of the similarity class $\tilde{[x_i]_B}$, the cardinality of similarity class $\tilde{[x_i]_B}$ can be calculated with $|\tilde{[x_i]_B}| = \sum_{j=1}^n r_{ij}$. r_{ij} is the similar relation value between samples x_i and x_j , the value of r_{ij} in $[0, 1]$, which is a major difference from the equivalence relation. Furthermore, when $r_{ij} \in \{0, 1\}$, fuzzy information entropy degenerates into the case of classical Shannon entropy [36]. The value of $H(\tilde{B})$ indicates the quantity of information contained in the feature subset B .

Hu et al. [38] have also presented forms of fuzzy information entropy measures for binary features, such as fuzzy joint entropy, fuzzy conditional entropy, and joint conditional entropy, used to measure the relationship between features.

Definition 2. [38] Given an $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, The fuzzy joint entropy E to B is defined as:

$$H(\tilde{E}, \tilde{B}) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|\tilde{[x_i]_B} \cap \tilde{[x_i]_E}|}{n}, \quad (2)$$

where $\tilde{[x_i]_B}$ and $\tilde{[x_i]_E}$ respectively represent the similarity classes generated for the sample x_i under the feature subsets B and E . The joint conditional entropy $H(\tilde{E}, \tilde{B})$ is represented as the quantity of information contained jointly by the feature subset E and the feature subset B .

Definition 3. [38] Given an $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, the fuzzy conditional entropy E to B is defined as:

$$H(\tilde{E}|\tilde{B}) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|\tilde{[x_i]_B} \cap \tilde{[x_i]_E}|}{|\tilde{[x_i]_B}|}. \quad (3)$$

The fuzzy conditional entropy $H(\tilde{E}|\tilde{B})$ indicates the uncertainty of E when B is known. The smaller the value of the fuzzy conditional entropy, the less uncertainty there is about E when B is already known.

Definition 4. [38] Given an $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, the fuzzy mutual information is given as follows:

$$I(\tilde{E}; \tilde{B}) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|\tilde{[x_i]_B}| \cdot |\tilde{[x_i]_E}|}{n \cdot |\tilde{[x_i]_B} \cap \tilde{[x_i]_E}|}. \quad (4)$$

B. Symmetric Uncertainty

Mutual information is commonly used to measure feature significance. However, relying solely on mutual information fails to distinguish between two features with the same quantity of mutual information. Therefore, Witten et al. [43] proposed symmetric uncertainty to measure the significance of features.

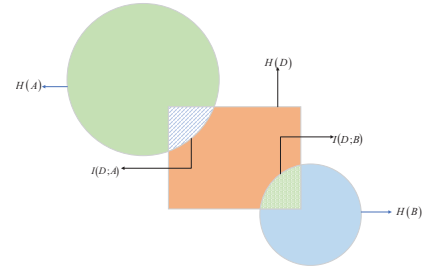


Fig. 1: Symmetric uncertainty illustration figure.

Definition 5. [43] Given an $FDT = \langle U, C \cup D, V \rangle$, for any feature subset $B \subseteq C$, the definition of symmetric uncertainty is defined as:

$$SU(D; B) = \frac{2I(D; B)}{H(\tilde{B}) + H(D)}. \quad (5)$$

In Fig 1, the two feature subsets A and B have the same quantity of mutual information with the decision D . However, it is evident that feature set A contains a large amount of irrelevant and redundant information, while B has relatively less redundancy and better avoids information loss. Therefore, considering only mutual information is insufficient.

III. MACROGRANULAR ENTROPY MEASUREMENT

In this section, we point out the limitations of fuzzy information entropy. We introduce the concept of macrogranular entropy and its generalized form for measuring relationships between features, highlighting the advantages of macrogranular entropy over fuzzy information entropy. Since symmetric uncertainty does not account for redundancy between features, we further introduce the concept of the degree of symmetric association, which considers both feature relevance and redundancy between features. Additionally, we propose a feature selection algorithm based on the degree of symmetric association.

A. Macrogranular Entropy

Definition 6. Given an $FDT = \langle U, C \cup D, V \rangle$, for a feature subset $B \subseteq C$, then the macrogranular entropy of B is defined as:

$$\widetilde{MH}(B) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|\widetilde{[x_i]_B}|}{n}, \quad (6)$$

where $|\widetilde{[x_i]_B}|$ is the cardinality of fuzzy similarity class $\widetilde{[x_i]_B}$. The measurement of a single feature's information quantity is consistent with the fundamental form of fuzzy information entropy as Definition 1. Macrogranular Entropy and Fuzzy Information Entropy share the same fundamental form, both derived from Shannon entropy, and characterize the amount of information in features from a probabilistic perspective. When the fuzzy relation degenerates into an equivalence relation, their forms become consistent with that of Shannon entropy. The main difference between macrogranular entropy and fuzzy entropy lies in the measurement of information among multiple features.

Definition 7. Given an $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, the joint macrogranular entropy of B and E is defined as:

$$\widetilde{MH}(B, E) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_B}| \cdot |\widetilde{[x_i]_E}|}{n |\widetilde{[x_i]_B \cup [x_i]_E}|} \right), \quad (7)$$

where $|\widetilde{[x_i]_B \cup [x_i]_E}| = \sum_{x_i \in U} \max \{ \mathfrak{R}_B(x_i, x_j), \mathfrak{R}_E(x_i, x_j) \}$, $|\widetilde{[x_i]_B}|$ and $|\widetilde{[x_i]_E}|$ represent the similarity class generated by x_i under the feature subsets B and E , $|\bullet|$ represents the cardinality of the similarity class.

The joint macrogranular entropy we propose differs from form of fuzzy joint entropy as in Definition 2. Fuzzy joint entropy involves a simple intersection operation, but different features may have different properties. Therefore, we cannot perform simple intersection operations on similarity relations generated from different features.

Proposition 1. Given an information table $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, if $B \subseteq E$ we have $\widetilde{MH}(B, E) = \widetilde{MH}(E) = H(\tilde{E})$.

Proof. Assume that $\mathfrak{R}_B, \mathfrak{R}_E$ are similarity relations generated by B, E , because of $B \subseteq E, \forall x_i, x_j \in U$ means

that B has a rougher fuzzy similarity class division. We can obtain $\mathfrak{R}_B(x_i, x_j) \geq \mathfrak{R}_E(x_i, x_j)$ means similarity class taking the union $|\widetilde{[x_i]_B}|$ and $|\widetilde{[x_i]_E}|$ results in $|\widetilde{[x_i]_B \cup [x_i]_E}| = |\widetilde{[x_i]_B}|$ then $|\widetilde{[x_i]_B \cup [x_i]_E}| = |\widetilde{[x_i]_B}|$, in $\widetilde{MH}(B, E) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_B}| \cdot |\widetilde{[x_i]_E}|}{n |\widetilde{[x_i]_B \cup [x_i]_E}|} \right)$ we can simultaneously simplify the numerator and denominator by canceling out one instance of $|\widetilde{[x_i]_B}|$ therefore $\widetilde{MH}(B, E) = \widetilde{MH}(E) = H(\tilde{E})$. $\square$

Proposition 1 indicates that when B is a subset of E , E contains all the information of B , then the joint macrogranular entropy of E and B equals the fuzzy information entropy of E .

Proposition 2. Given a fuzzy decision table $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, we have $\widetilde{MH}(B, E) \geq \max \{ \widetilde{MH}(E), \widetilde{MH}(B) \}$.

Proof. Assume that $\mathfrak{R}_B, \mathfrak{R}_E$ are fuzzy relations generated by B, E , we can obtain fuzzy similarity class taking the union $|\widetilde{[x_i]_B \cup [x_i]_E}| \geq |\widetilde{[x_i]_B}|$ and $|\widetilde{[x_i]_B \cup [x_i]_E}| \geq |\widetilde{[x_i]_E}|$ then we have $1 \max \left\{ \frac{|\widetilde{[x_i]_B}|}{|\widetilde{[x_i]_B \cup [x_i]_E}|}, \frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_B \cup [x_i]_E}|} \right\} \leq 1$, so we can get $\frac{|\widetilde{[x_i]_B}| \cdot |\widetilde{[x_i]_E}|}{n |\widetilde{[x_i]_B \cup [x_i]_E}|} \leq \left\{ \frac{|\widetilde{[x_i]_E}|}{n}, \frac{|\widetilde{[x_i]_B}|}{n} \right\}$ therefore $\widetilde{MH}(B, E) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_B}| \cdot |\widetilde{[x_i]_E}|}{n |\widetilde{[x_i]_B \cup [x_i]_E}|} \right) \geq \max \left\{ -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E}|}{n} \right), -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_B}|}{n} \right) \right\}$. Proposition 2 obtains the proof. $\square$

Proposition 2 indicates that the value of the joint macrogranular entropy of any two feature subsets is not less than the macrogranular entropy value of either of the individual feature subsets. Information quantity of the union of any two feature subsets is greater than or equal to the information content of either feature subset alone.

Proposition 3. Given an $FDT = \langle U, C \cup D, V \rangle$, for feature subsets $B, E \subseteq C$ we have $\widetilde{MH}(B, E) \geq 0$.

Proof. The proof can be easily obtained from proof of Proposition 2 and Definition 1. $\square$

Proposition 3 indicates that macrogranular entropy measurement possesses the non-negativity.

Definition 8. Given an $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, the form of conditional macrogranular entropy of E when B is defined as:

$$\widetilde{MH}(E|B) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_E \cup [x_i]_B}|} \right). \quad (8)$$

$\widetilde{MH}(E|B)$ is used to measure the uncertainty of E when B is known. The smaller the value of $\widetilde{MH}(E|B)$, the more information about E is contained in B , and thus the smaller uncertainty of E .

Proposition 4. Given an $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, if $E \subseteq B$, then $\widetilde{MH}(E|B) = 0$.

Proof. Assume that $\mathcal{R}_B, \mathcal{R}_E$ are similarity relations generated by B, E , because of $E \subseteq B$ means that E has a rougher fuzzy similarity class division. $\forall x_i, x_j \in U$ we have $\mathcal{R}_B(x_i, x_j) \leq \mathcal{R}_E(x_i, x_j)$. So $\widetilde{[x_i]_B} \cup \widetilde{[x_i]_E} = \max\{\widetilde{[x_i]_B}, \widetilde{[x_i]_E}\} = \widetilde{[x_i]_E}$ then $|\widetilde{[x_i]_B} \cup \widetilde{[x_i]_E}| = |\widetilde{[x_i]_E}|$, we can obtain $\frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_B} \cup \widetilde{[x_i]_E}|} = 1$, then $\widetilde{MH}(E|B) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_B} \cup \widetilde{[x_i]_E}|} \right) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_E}|} \right) = 0$ proposition is proved. $\square$

Since the meaning of conditional macrogranular entropy is to measure the uncertainty, Proposition 4 indicates that when the information quantity of feature subset is contained by feature subset B , the conditional macrogranular entropy of E relative to B is 0, meaning uncertainty of E is 0.

Proposition 5. Given an $FDT = \langle U, C \cup D, V \rangle$, for a feature subset $B \subseteq C$ and a conditional feature $a_i \in C$, we have $\widetilde{MH}(E|B) \geq \widetilde{MH}(E|B \cup a_i)$.

Proof. Assume that $\mathcal{R}_B$ is the similarity relation generated by B . For any fuzzy similarity relation $\mathcal{R}(x_i, x_j)$ between samples x_i and x_j , $B \cup a_i$ has a finer fuzzy similarity class division. We have $\mathcal{R}_B(x_i, x_j) \geq \mathcal{R}_{B \cup a_i}(x_i, x_j)$, therefore $|\widetilde{[x_i]_B}| \geq |\widetilde{[x_i]_{B \cup a_i}}|$, we can easily get $|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_B}| \geq |\widetilde{[x_i]_E} \cup \widetilde{[x_i]_{B \cup a_i}}|$, so we have $\frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_B}|} \leq \frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_{B \cup a_i}}|}$. Thus $-\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_B}|} \right) \geq -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_{B \cup a_i}}|} \right)$. The proposition is proved. $\square$

Proposition 5 indicates that conditional macrogranular entropy is monotonic.

Proposition 6. Given an $FDT = \langle U, C \cup D, V \rangle$, for feature subsets $B, E \subseteq C$, if $B \subseteq E$ we have $\widetilde{MH}(E|B) = \widetilde{MH}(E) - \widetilde{MH}(B)$.

Proof. According to the forms in Definition 6 $\widetilde{MH}(B) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|\widetilde{[x_i]_B}|}{n}$ and $\widetilde{MH}(E) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|\widetilde{[x_i]_E}|}{n}$. According to Definition 8, since $B \subseteq E$, in the $\widetilde{MH}(E|B) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_B}|} \right)$, we have $|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_B}| = |\widetilde{[x_i]_B}|$. Thus, we obtain $\widetilde{MH}(E|B) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_B}|} \right)$, is equal to $\widetilde{MH}(E) - \widetilde{MH}(B)$. The proposition is proved. $\square$

Proposition 6 indicates that when B is contained by E , the uncertainty of E relative to B is the difference between the information quantity of E and B .

Definition 9. Given an $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, the mutual macrogranular information between B and E is defined as:

$$\widetilde{MMH}(B; E) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_B} \cup \widetilde{[x_i]_E}|}{n} \right). \quad (9)$$

The mutual macrogranular information $\widetilde{MMH}(B; E)$ can be used to measure the quantity of relevance between feature subsets B and E .

Proposition 7. Given an $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, if $B \subseteq E$ then $\widetilde{MMH}(E; B) = \widetilde{MH}(B)$.

Proof. Assume that $\mathcal{R}_B, \mathcal{R}_E$ are similarity relations generated by B, E , because of $B \subseteq E$, $\forall x_i, x_j \in U$ we can obtain $\mathcal{R}_B(x_i, x_j) \geq \mathcal{R}_E(x_i, x_j)$ means similarity class $\widetilde{[x_i]_B} \cup \widetilde{[x_i]_E} = \widetilde{[x_i]_B}$ then $|\widetilde{[x_i]_B} \cup \widetilde{[x_i]_E}| = |\widetilde{[x_i]_B}|$, therefore $\widetilde{MMH}(E; B) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_B} \cup \widetilde{[x_i]_E}|}{n} \right) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_B}|}{n} \right) = \widetilde{MH}(B)$. $\square$

Proposition 7 means that for any two feature subsets $B, E \subseteq C$, if $B \subseteq E$ the mutual macrogranular information is equivalent to the information quantity of B .

Proposition 8. Given an $FDT = \langle U, C \cup D, V \rangle$, for any two feature subsets $B, E \subseteq C$, the mutual macrogranular information $\widetilde{MMH}(E; B) \leq \min\{\widetilde{MH}(B), \widetilde{MH}(E)\}$.

Proof. Assume that $\mathcal{R}_B, \mathcal{R}_E$ are the similarity relations generated by feature subsets B and E . According to $\widetilde{MMH}(E; B) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_B}|}{n} \right)$, where $|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_B}|$ implies $|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_B}| \geq \max\{|\widetilde{[x_i]_B}|, |\widetilde{[x_i]_E}|\}$, then it follows that $\frac{|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_B}|}{n} \geq \max\left\{\frac{|\widetilde{[x_i]_E}|}{n}, \frac{|\widetilde{[x_i]_B}|}{n}\right\}$. According to Definition 6, we can derive $-\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E} \cup \widetilde{[x_i]_B}|}{n} \right) \leq \left\{ -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_B}|}{n} \right), -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E}|}{n} \right) \right\}$. Therefore, $\widetilde{MMH}(E; B) \leq \min\{\widetilde{MH}(B), \widetilde{MH}(E)\}$ is proved. $\square$

Proposition 8 indicates that for any two feature subsets, the mutual information between them does not exceed the quantity of information of any feature subset.

Based on Definitions 7, 8 and 9, for two feature subsets $B, E \subseteq C$, we can derive the following relationships between macrogranular entropy and the mutual macrogranular information, joint macrogranular entropy, and conditional macrogranular entropy:

- (1) $\widetilde{MMH}(B; E) = \widetilde{MH}(E) - \widetilde{MH}(E|B)$;
- (2) $\widetilde{MMH}(B; E) = \widetilde{MH}(B) - \widetilde{MH}(B|E)$;
- (3) $\widetilde{MH}(E|B) = \widetilde{MH}(B, E) - \widetilde{MH}(B)$;
- (4) $\widetilde{MH}(B, E) = \widetilde{MH}(E) + \widetilde{MH}(B) - \widetilde{MMH}(B; E)$.

In some environments, the generalized form of fuzzy information entropy presents computational challenges that hinder its applicability. In contrast, macrogranular entropy effectively circumvents these issues. To illustrate this distinction, we present Example 2, which clearly demonstrates the limitations of the fuzzy information entropy.

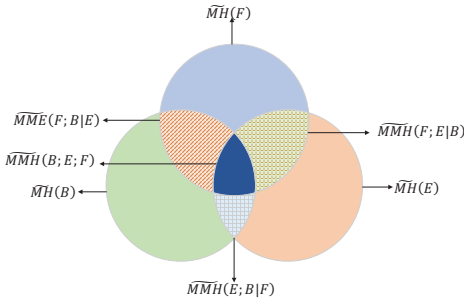


Fig. 2: The relationships among B, E, F .

Definition 10. Given an $FDT = \langle U, C \cup D, V \rangle$, for feature subsets B, E and $F \subseteq C$, when B and F are known, the definition of the joint conditional macrogranular entropy E relative to B and F is defined as:

$$\widetilde{MH}(E|B, F) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E}|}{|\widetilde{[x_i]_E \cup [x_i]_{B \cup F}}|} \right), \quad (10)$$

where $\widetilde{[x_i]_{B \cup F}} = \widetilde{[x_i]_B} \cap \widetilde{[x_i]_F}$, B and F in this formula are interchangeable.

Proposition 9. Given an $FDT = \langle U, C \cup D, V \rangle$, for feature subsets $B, E, F \subseteq C$, if $E \subseteq F$, $E \subseteq B$, or $E \subseteq B \cup F$, $\widetilde{MH}(E|B, F) = 0$.

Proof. Since $E \subseteq F$ or $E \subseteq B$, we can conclude that $E \subseteq F \cup B$. Thus, we have $\widetilde{[x_i]_{B \cup F}} \leq \widetilde{[x_i]_E}$, which leads to $\widetilde{[x_i]_E \cup [x_i]_{B \cup F}} = \widetilde{[x_i]_E}$. This implies that $\left| \frac{\widetilde{[x_i]_E}}{\widetilde{[x_i]_E \cup [x_i]_{B \cup F}}} \right| = 1$, therefore, $\widetilde{MMH}(E; F|B) = 0$. $\square$

Proposition 9 indicates that when E is a subset of B, E , or $B \cup F$, the uncertainty information of E with respect to B, E is 0.

Definition 11. Given an $FDT = \langle U, C \cup D, V \rangle$, for feature subsets B, E and $F \subseteq C$, when B and F are known, the definition of the joint mutual macrogranular information B relative to E and F is defined as:

$$\widetilde{MMH}(E, F; B) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_B \cup [x_i]_{E \cup F}}|}{n} \right), \quad (11)$$

where $\widetilde{[x_i]_{E \cup F}} = \widetilde{[x_i]_E} \cap \widetilde{[x_i]_F}$, B and F in this formula are interchangeable.

Proposition 10. Given an $FDT = \langle U, C \cup D, V \rangle$, for feature subsets $B, E, F \subseteq C$, if $B \subseteq F$, $B \subseteq E$, or $B \subseteq E \cup F$, $\widetilde{MH}(E, F; B) = \widetilde{MH}(B) = H(B)$.

Proof. It can be easily obtained from proof of Proposition 7. $\square$

Proposition 10 indicates that when B is a subset of E, F , or $E \cup F$, the joint mutual macrogranular information with B and E, F is equal to the information quantity of B .

Based on Definitions 1, 7, 10, and 11, we can establish that macrogranular entropy, joint mutual macrogranular information, and joint conditional macrogranular entropy have the following relationships:

- (1) $\widetilde{MMH}(E, F; B) = \widetilde{MH}(B) - \widetilde{MH}(B|E, F)$;
- (2) $\widetilde{MMH}(E, B; F) = \widetilde{MH}(F) - \widetilde{MH}(F|E, B)$;
- (3) $\widetilde{MMH}(B, F; E) = \widetilde{MH}(E) - \widetilde{MH}(E|B, F)$.

Definition 12. Given an $FDT = \langle U, C \cup D, V \rangle$, for feature subsets B, E and $F \subseteq C$, the conditional mutual macrogranular information between E and F relative to B when B is defined as:

$$\widetilde{MMH}(E; F|B) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\widetilde{[x_i]_E \cup [x_i]_{B \cup F}}|}{|\widetilde{[x_i]_E \cup [x_i]_B}|} \right), \quad (12)$$

where E and F can not be interchanged, because this formula describes the mutual macrogranular information between the uncertainty information of F relative to B and E .

The conditional macrogranular mutual information $\widetilde{MMH}(E; F|B)$ can be used to measure the degree of relevance of the unknown parts between feature subsets E and F when the feature subset B is known, specifically the relevance between the information in E and F that is not contained within B as shown in Fig 2.

Proposition 11. Given an $FDT = \langle U, C \cup D, V \rangle$, for feature subsets B, E and $E \subseteq C$, if $E, F \subseteq B$ we have $\widetilde{MMH}(E; F|B) = 0$.

Proof. Because $E, F \subseteq B$ we can obtain $|\widetilde{[x_i]_B}| \leq \min \{ |\widetilde{[x_i]_E}|, |\widetilde{[x_i]_F}| \}$. Therefore $|\widetilde{[x_i]_{B \cup F}}| = |\widetilde{[x_i]_B}|$ and $|\widetilde{[x_i]_E \cup [x_i]_B}| = |\widetilde{[x_i]_E}|$, then we can conclude that $\left| \frac{|\widetilde{[x_i]_E \cup [x_i]_{B \cup F}}|}{|\widetilde{[x_i]_E \cup [x_i]_B}|} \right| = 1$, then $\widetilde{MMH}(E; F|B) = 0$. $\square$

Proposition 11 indicates that when E and F are subsets of B , the relevance of the unknown information between them is 0, meaning B contains all the information of E and F , as can be derived from Fig 2.

Proposition 12. Given an $FDT = \langle U, C \cup D, V \rangle$, for feature subsets B, E and $E \subseteq C$, $\widetilde{MMH}(E; F|B) \geq 0$.

Based on Definitions 8, 10, and 12, we can establish that conditional macrogranular entropy, conditional mutual macrogranular information, and joint conditional macrogranular entropy have the following relationships:

- (1) $\widetilde{MMH}(F; E|B) = \widetilde{MH}(E|B) - \widetilde{MH}(E|B, F)$;
- (2) $\widetilde{MMH}(E; F|B) = \widetilde{MH}(F|B) - \widetilde{MH}(F|B, E)$.

B. Comparison between Macrogranular Entropy and Fuzzy Information Entropy

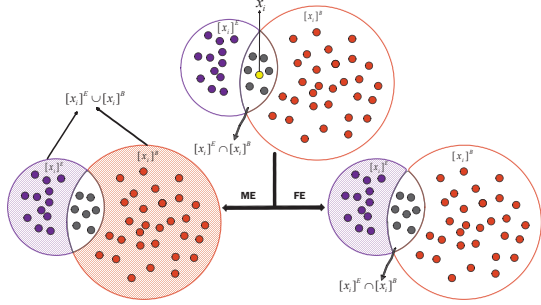


Fig. 3: Comparison of macrogranular entropy and fuzzy information entropy.

In this section, we will further elaborate on why macrogranular entropy outperforms fuzzy entropy in measuring relationships between features within fuzzy sample relations from a more detailed perspective. First, we present an illustrative figure, as shown in Fig 3. In this figure, the yellow sample x_i belongs to the red fuzzy similarity class generated by the feature subset B and the purple fuzzy similarity class generated by the feature subset E .

We first illustrate the differences between fuzzy conditional entropy and conditional macrogranular entropy by analyzing their respective definitions. According to the definition of fuzzy conditional entropy $H(\tilde{E}|\tilde{B}) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\tilde{x}_i|_{\tilde{E}} \cap \tilde{x}_i|_{\tilde{B}}|}{|\tilde{x}_i|_{\tilde{B}}|} \right)$ as shown in the right of Fig 3 (FE), it only considers the relationship between the overlapping black region and the purple region, while the red region is ignored. On the other hand, conditional macrogranular entropy is defined as $\widetilde{MH}(E|B) = -\frac{1}{n} \sum_{i=1}^n \left(\log \frac{|\tilde{x}_i|_{\tilde{E}}|}{|\tilde{x}_i|_{\tilde{E}} \cup \tilde{x}_i|_{\tilde{B}}|} \right)$ which takes into account not only the overlapping black region but also the red and purple regions through the union operation $|\tilde{x}_i|_{\tilde{E}} \cup \tilde{x}_i|_{\tilde{B}}|$. Through our research, we found that during the feature selection process, only the feature space changes while the decision remains fixed. Therefore, in Macrogranular Entropy framework, the union operation treats the decision-based partitioning of the sample space as a lower bound, focusing on the changes induced by features in the sample space. This approach avoids the limitation in existing information measurement frameworks, where the intersection operation constrains feature-induced variations as the lower bound.

For further illustrate the differences between the conditional macrogranular entropy and the fuzzy conditional entropy, we provide the following examples for specific explanation.

Example 1. Given an $FDT = \langle U, C \cup D, V \rangle$ as shown in Table I, for any two feature $a_1 \subseteq C$ and $a_2 \subseteq C$. The fuzzy similarity relationship matrices generated by a_1 , $a_1 \cup a_2$, and equivalence relation matrix generated by the decision D are

TABLE I: Description of fuzzy decision table.

U	a_1	a_2	a_3	D
x_1	0.820	0.8225	1	1
x_2	0.1344	0.9757	0.2500	0
x_3	0.8334	0.0125	0.2400	1
x_4	0.8328	0.3420	0.8800	1
x_5	0.8125	0.0554	0.5700	0
x_6	0.3452	0.2482	0.5500	0

as follows:

$$\mathcal{R}_{a_1} = \begin{bmatrix} 1 & 0.0850 & 0.9993 & 0.9994 & 0.9995 & 0.3056 \\ 0.0850 & 1 & 0.0782 & 0.0786 & 0.0909 & 0.7932 \\ 0.9993 & 0.0782 & 1 & 1 & 0.9977 & 0.2885 \\ 0.9994 & 0.0786 & 1 & 1 & 0.9979 & 0.2894 \\ 0.9995 & 0.0909 & 0.9977 & 0.9979 & 1 & 0.3202 \\ 0.3056 & 0.7931 & 0.2885 & 0.2894 & 0.3202 & 1 \end{bmatrix},$$

$$\mathcal{R}_{a_1 \cup a_2} = \begin{bmatrix} 1 & 0.0850 & 0.1303 & 0.4881 & 0.1607 & 0.3056 \\ 0.0850 & 1 & 0.0560 & 0.0786 & 0.0720 & 0.1932 \\ 0.1303 & 0.0560 & 1 & 0.7137 & 0.9943 & 0.2885 \\ 0.4881 & 0.0786 & 0.7137 & 1 & 0.7748 & 0.2894 \\ 0.1607 & 0.0720 & 0.9943 & 0.7748 & 1 & 0.3202 \\ 0.3056 & 0.1932 & 0.2885 & 0.2894 & 0.3202 & 1 \end{bmatrix},$$

$$\mathcal{R}_D = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \end{bmatrix}.$$

According to Equation 4, the fuzzy conditional entropy, and conditional macrogranular entropy as Equation 8, the calculation results are as follows: $H(D|a_1) = 0.6588$ and $H(D|a_1 \cup a_2) = 0.6604$. It is clearly evident that the conditional entropy for measuring the relationship between feature subset and the decision is not monotonic. However, based on conditional macrogranular entropy, we can calculate that $\widetilde{MH}(D|a_1) = 0.5191$, $\widetilde{MH}(D|a_1 \cup a_2) = 0.4054$. Clearly, the conditional macrogranular entropy proposed by us is monotonic.

Example 2. For a sample set $X = \{x_1, x_2, x_3\}$, if we have two relationships $\mathcal{R}_i, \mathcal{R}_j$ generated on X based features a_i and a_j are as follows:

$$\mathcal{R}_i = \begin{bmatrix} 0.1 & 0.2 & 0 \\ 0.4 & 0 & 0.5 \\ 1 & 0 & 0 \end{bmatrix}, \quad \mathcal{R}_j = \begin{bmatrix} 0 & 0 & 0.8 \\ 0 & 1 & 0.5 \\ 0 & 0.5 & 0.4 \end{bmatrix}$$

According to the Definitions 2 to 4, fuzzy conditional entropy $H(\mathcal{R}_i|\mathcal{R}_j) = \odot$, fuzzy joint entropy $H(\mathcal{R}_i, \mathcal{R}_j) = \odot$, and fuzzy mutual information $I(\mathcal{R}_i; \mathcal{R}_j) = \odot$. $\odot$ indicates that the result is meaningless or cannot be computed. However, based on the macrogranular entropy form we provided in Definitions 7 to 9, we can calculate that $\widetilde{MH}(\mathcal{R}_i|\mathcal{R}_j) = 1.2928$, $\widetilde{MH}(\mathcal{R}_i, \mathcal{R}_j) = 2.8408$ and $\widetilde{MMH}(\mathcal{R}_i; \mathcal{R}_j) = 0.9218$.

This demonstrates that macrogranular entropy for measuring the relationships between features can be applied in more complex sample relationships.

C. Feature Selection Based on Macrogranular Entropy.

Symmetric uncertainty lacks consideration of the redundancy between selected features and candidate feature, focusing solely on the relevance between candidate feature and the decision. We propose a novel concept that the degree of symmetric association is inspired by symmetric uncertainty. Information redundancy between features can reduce the generalization ability of the final feature subset. We introduce the concept of the degree of symmetric association, which transforms correlation and redundancy into a ratio-based form for consideration.

Definition 13. Given an $FDT = \langle U, C \cup D, V \rangle$, $\forall a_i \in C$, $\forall a_j \in S$, S is the already selected feature subset. The degree of symmetric association is defined as: $SIG(a_i) =$

$$DSA(a_i) = \frac{\sum_{a_s \in S} (\widetilde{MMH}(D; a_s | a_i) + \widetilde{MMH}(D; a_i | a_s))}{|S| \widetilde{MH}(D, a_i) + \sum_{a_s \in S} \widetilde{MH}(D, a_s)}, \quad (13)$$

where S represents the current selected feature subsets, a_i is the feature under consideration for significance evaluation, and a higher computed value indicates greater feature significance. Additionally, if the current feature subset space $S = \emptyset$, then $SIG(a_i) = \widetilde{MMH}(a_i; D)$.

The degree of symmetric association considers not only the conditional mutual information of selected feature a_s and candidate feature a_i , but also their joint information together. This is a more comprehensive approach to measure the significance of the candidate features.

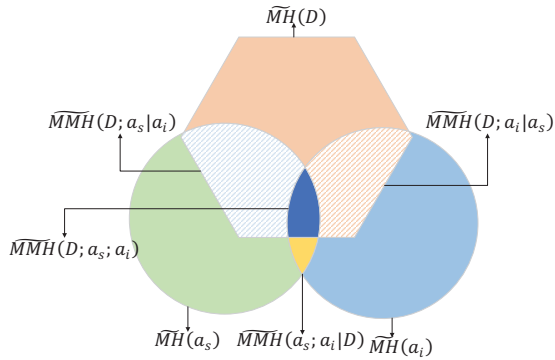


Fig. 4: The figure of the degree of symmetrical association.

As illustrated in the Fig.4, relying solely on mutual information measures overlooks the information redundancy and interaction between features. Since the information among features in the feature space is often intertwined, mutual information alone is insufficient for effective feature selection. Due to feature redundancy in high-dimensional real-world datasets, ADSA algorithm is designed to reduce redundancy while considering mutual information with the decision. To demonstrate this effect, we randomly selected two features from the original dataset and compared them with two features selected by ADSA. Scatter plots in Fig. 5 clearly show that ADSA outperforms the randomly selected features in discriminating between decision classes.

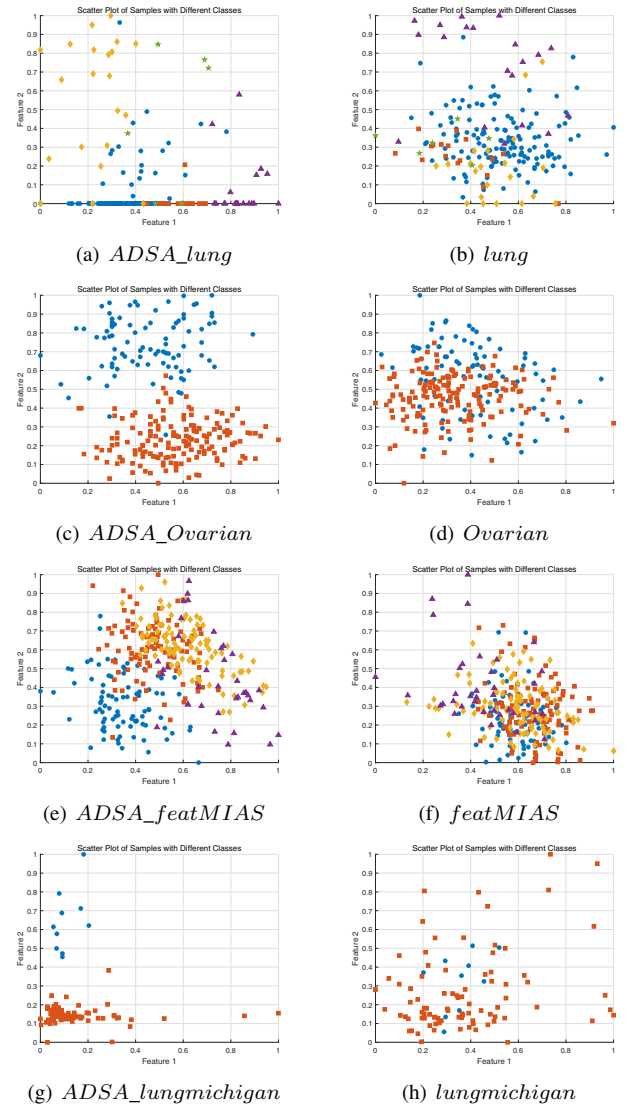


Fig. 5: Top two features by ADSA and two random features of the original datasets.

We propose an algorithm called feature selection algorithm based on the degree of symmetric association (ADSA) as shown in Fig 6. We have previously demonstrated that conditional macrogranular entropy is monotonic, so we use the change in conditional macrogranular entropy as a stopping criterion. We also propose a feature selection algorithm based on monotonic conditional macrogranular entropy (MCME).

Definition 14. The significance of candidate feature a_i is defined as:

$$SIG(a_i) = \widetilde{MH}(D|red) - \widetilde{MH}(D|red \cup a_i), \quad (14)$$

where red is the current selected feature subset.

For each feature $a_i \in C$, the algorithm computes the similar relation R_{a_i} (line 3), which requires comparing all sample pairs. The time complexity for this step is $O(Mn^2)$. The calculation of $\widetilde{MH}(D|C)$ (line 5) depends on the entire feature set C , contributing at most $O(n^2)$ to the overall

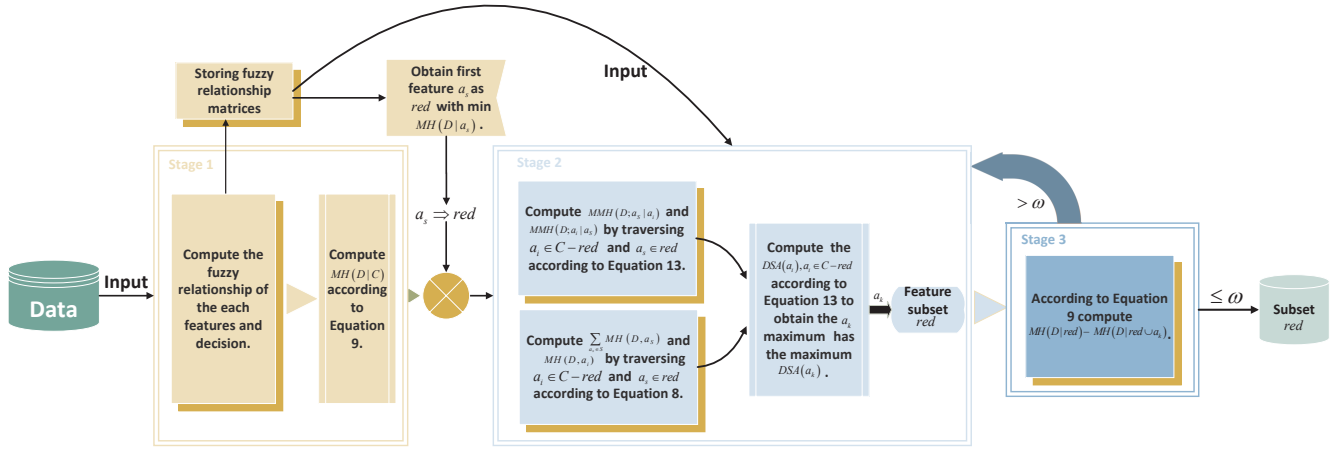


Fig. 6: The flowchart of ADSA algorithm.

Algorithm 1 Feature selection algorithm based on the degree of symmetric association (ADSA).

Input: $FDT = \langle U, C \cup D, V \rangle$, parameter ω .

Output: Feature subset red .

```

1: Initialize  $red = \emptyset$ ;
2: for each  $a_i \in C$  do
3:   Compute the similarity relation  $R_{a_i}$  according to Equation 15;
4: end for
5: Compute the conditional macrogranular entropy  $\widetilde{MH}(D|C)$  according to Equation 8;
6: while  $\widetilde{MH}(D|red) \neq \widetilde{MH}(D|C)$  do
7:   for each  $a_i \in C - red$  do
8:     for each  $a_s \in red$  do
9:       Compute the cumulative sum  $\sum_{a_s \in S} (\widetilde{MMH}(D; a_s | a_i) + \widetilde{MMH}(D; a_i | a_s))$ , according to Equation 12;
10:      Compute the  $\widetilde{MMH}(D; a_s | a_i) + \widetilde{MMH}(D; a_i | a_s)$  according to Equation 7
11:    end for
12:    Compute the  $DSA(a_i)$  according to Equation 13;
13:  end for
14:  Find  $a_k \in C - red$ , where  $DSA(a_k)$  is the biggest value;
15:  if  $\widetilde{MH}(D|red) - \widetilde{MH}(D|red \cup a_k) > \omega$  then
16:     $red \leftarrow red \cup \{a_k\}$ ;
17:     $C \leftarrow C - a_k$ ;
18:  else
19:    return  $red$ ;
20:  end if
21: end while
22: return  $red$ 

```

complexity. For each candidate feature $a_i \in C - red$, the algorithm computes the cumulative sum and symmetric association degree $DSA(a_i)$ (lines 8-12). Each computation involves iterating over the features in red , leading to an

Algorithm 2 Forward greedy feature selection algorithm based on the monotonic conditional macrogranular entropy (MCME).

Input: $FDT = \langle U, C \cup D, V \rangle$, parameter ω .

Output: Feature subset red .

```

1: Initialize  $red = \emptyset$ ;
2: for each  $a_i \in C$  do
3:   Compute the similarity relation  $R_{a_i}$ ;
4: end for
5: Compute the conditional macrogranular entropy  $\widetilde{MH}(D|C)$  according to Equation 8;
6: while  $\widetilde{MH}(D|red) \neq \widetilde{MH}(D|C)$  do
7:   for each  $a_i \in C - red$  do
8:     Compute the  $SIG(a_i) = \widetilde{MH}(D|red) - \widetilde{MH}(D|red \cup a_i)$  according to Equation 14;
9:   end for
10:  Find  $a_k \in C - red$ , where the  $SIG(a_k)$  is the biggest value;
11:  if  $\widetilde{MH}(D|red) - \widetilde{MH}(D|red \cup a_k) > \omega$  then
12:     $red \leftarrow red \cup \{a_k\}$ ;
13:     $C \leftarrow C - a_k$ ;
14:  else
15:    return  $red$ ;
16:  end if
17: end while
18: return  $red$ 

```

approximate complexity of $O(n^2 \times k)$ per feature. Since this process is repeated for at most $M - k$ features, the overall complexity of this stage is $O(n^2 \times (M - k) \times k)$. Combining these steps, the dominant term in the total complexity of ADSA is $O(n^2 \times ((2M - k) \times k/2))$. All of data and code is publicly available at https://github.com/woshidagongzai/woshidagongzai-MacroG_Entropy_ADSA.

IV. EXPERIMENTAL ANALYSIS

In this section, we compare feature selection algorithm based on the monotonic conditional macrogranular entropy (MCME) and feature selection algorithm based on the degree of symmetric association (ADSA) with eight other state-of-the-art algorithms. Because the neighborhood rough set model is also an important feature selection tool, the eight other state-of-the-art algorithms include three based on neighborhood rough set model and the other three are based on fuzzy rough set model. These include an algorithm based on information gain ratio (Gain Ratio) [9], an algorithm based on fitting fuzzy rough set model (NFRS) [34], an algorithm based on fuzzy self-information (FSI) [44], an algorithm based on fuzzy conditional implication granularity (MFIGI) [16], an algorithm based on fuzzy conditional entropy [38], an forward heterogeneous feature selection based on neighborhood rough sets (NFARNRS) [45], an algorithm based on neighborhood self-information (NSI) [46], a feature selection algorithm based on neighborhood combination entropy (FScNCE) [26].

All experiments are implemented by MATLAB R2022a, and run on a personal computer with an Intel Core i9-12900H CPU at 5.00 GHz, 32.0 GB RAM.

A. Datasets

In experimental section, prior to feature selection, we first normalize the features as $F(x_i, a_k) = \frac{f(x_i, a_k) - \min_{a_k}}{\max_{a_k} - \min_{a_k}}$, where $\max_{a_k}$ represents the maximum value in feature a_k , $\min_{a_k}$ is the minimum value in feature a_k , and $f(x_i, a_k)$ is the feature value of sample x_i under feature a_k . For all datasets, we handle missing features as follows: For numerical data, we use the window median imputation method with a window size of 10. For categorical data, we apply forward imputation. The details of datasets used in paper are as shown in Table II. The datasets Pageblock, Solar-Flare, Seismic, Soybean, Splice, Isolet, Lung, and Lung-Michigan are from publicly available UCI datasets, including both text and numerical types. Yale is from the Yale Face Database at Yale University, and FeatMIAS and ORL are from Kaggle. Leukemia and Carcinom are from the NCBI GEO and SRA data portal. Nci9, Pixraw10P, Orlraws10P, Ovarian, and GLI-85 are from the Feature Selection project at Arizona State University on GitHub <https://jundongli.github.io/scikit-feature/>. A large portion of the datasets are derived from biology, genetics, or medical-related domains. In medical diagnostics, a significant number of examinations are often required to investigate the pathogenic factors of diseases. Feature selection can help uncover disease-related features, thereby improving treatment efficacy and disease prevention. Moreover, genetic data often involve extremely high dimensionality; dimensionality reduction through feature selection can aid in identifying relevant pathogenic genes, enhancing the interpretability of medical research. Additionally, datasets from the image domain are included. In this field, feature selection helps reduce the dimensionality of image data, which in turn accelerates model training and improves computational efficiency.

B. Evaluation metrics

We use SVM and NB classifiers for evaluating the quality of a selected feature subset. For a given feature subset, we can train an SVM or NB classifier using that subset and assess its classification accuracy on a validation or test set. Generally, the better the classification accuracy, the more effective the feature subset is. By comparing the F1-score and Recall of models trained with different feature subsets on the validation or test set, we can evaluate which subset is more effective. Generally, a feature subset that yields a higher F1-score or Recall is considered to be more optimal.

SVM: We used MATLAB's built-in SVM function `fitcsvm` to call the SVM classifier. Internally, we applied kernel function analysis, inputting the selected features and decision. The model was evaluated using ten-fold cross-validation, returning the accuracy and the standard deviation of accuracy.

NB: We used MATLAB's built-in Naive Bayes function `fitcnb` to call the Naive Bayes classifier. Internally, we applied probability distribution analysis (Gaussian, Kernel), inputting the selected features and decision. The model was evaluated using ten-fold cross-validation, returning the accuracy and the standard deviation of accuracy.

Recall : A metric used to evaluate the ability of the SVM classification model to correctly identify positive instances. It is calculated as the ratio of true positive predictions to the total number of actual positive instances. For SVM model, recall is computed using the formula: $\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$. Here, True Positives (TP) refers to the number of positive samples correctly predicted as positive by the SVM model, and False Negatives (FN) refers to the number of positive samples incorrectly predicted as negative by the SVM model.

F1Score: A metric that combines precision and recall into a single value, providing a balanced measure of the SVM performance, especially in imbalanced datasets. It ranges from 0 to 1, where a higher F1 Score indicates a better balance between precision and recall. For SVM model, the F1 Score is calculated using the formula: $F1\text{Score} = 2 \times \frac{\text{Accuracy} \times \text{Recall}}{\text{Accuracy} + \text{Recall}}$.

C. Parameter description

Our algorithms (ADSA) and (MCME) are heuristic algorithms, especially the MCME algorithm being a monotonic feature selection algorithm, we have set a stopping parameter ω for the feature selection process. For high-dimensional datasets, we set $\omega = 0.01$, and for low-dimensional datasets, we set it to 0.001. In this paper, high-dimensional datasets have more than 1000 features, while low-dimensional datasets with feature dimensions range from 0 to 1000.

Since macrogranular entropy measurement is based on the fuzzy rough set model, we need to construct fuzzy similarity relationships between samples. We calculate the fuzzy similarity relationship values between samples using the following form.

For feature a_k , the fuzzy similarity value between two samples x_i and x_j is computed by gaussian kernel [47]:

$$r_{x_i, x_j}^{a_k} = \exp\left(-\frac{|x_i - x_j|_{a_k}^2}{2\delta^2}\right), \quad (15)$$

TABLE II: Description of datasets.

No.	Dataset	Samples	Features	Classes	Feature Type	Area
1	Pageblock	5473	10	5	Integer, Continuous	Computer Science
2	Solar-Flare	1066	12	3	Categorical	Physics and Chemistry
3	Seismic	2584	18	2	Categorical, Continuous	Other
4	Soybean	683	35	19	Categorical	Biology
5	Splice	3190	60	3	Categorical	Biology
6	Yale	1024	165	15	Continuous	Image
7	FeatMIAS	322	280	4	Continuous	Medical Image
8	Isolet	1560	617	26	Continuous	Spoken letter recognition
9	ORL	400	1024	40	Continuous	Image
10	Lung	203	3312	4	Continuous	Medical
11	Leukemia	64	4702	2	Discrete	Biology
12	Lung-Michigan	96	7129	2	Continuous	Medical
13	Carcinom	174	9182	11	Continuous	Biology
14	Nci9	60	9712	9	Discrete	Biology
15	Pixraw 10P	100	10000	10	Continuous	Image
16	Orlraws 10P	100	10304	10	Continuous	Image
17	Ovarian	253	15154	2	Continuous	Biology
18	GLI-85	85	22283	2	Continuous	Biology
AVG.		968.72	5222.17	9.39		

where δ represents the standard deviation of the current feature a_k , and $|x_i - x_j|_{a_k}^2$ denotes the squared Euclidean distance between samples x_i and x_j under feature a_k .

The feature selection process stops if the evaluation of the features for the selected feature subset S results in a reduction of the uncertainty value of the decision D less than ω . Thus, the value of the conditional macrogranular entropy of D with respect to selected feature subset S decreases less than ω .

D. Analysis of Experimental Results

NFARNRS, NSI, and FScNCE are based on the neighborhood rough set model, which includes a neighborhood radius parameter, we iteratively select neighborhood radius sizes ranging from 0.05 to 0.3 with a step size of 0.05 in the feature selection algorithms. The feature subset that optimizes the metrics is then chosen as the final result for comparison.

Tables III to V present a comparison of the feature subset evaluation results of our algorithm and eight other algorithms under different metrics.

On the NB classifier as shown in Table III. The ADSA algorithm achieved the best performance on 9 datasets, and on average, its classification accuracy exceeded that of the original datasets by 12.22%, compared to 10.07% for the MCME algorithm. Both ADSA and MCME are effective in eliminating redundant information from the data, thereby enhancing the classification performance of the models.

On the SVM classifier as shown in Table IV. The ADSA algorithm achieved first or second place rankings on 16 datasets, while the MCME algorithm ranked first or second on 11 datasets. In terms of average accuracy, ADSA reached 88.25%, representing an improvement of 2.88% over the original dataset accuracy of 85.37%. Meanwhile, MCME achieved an average accuracy of 86.38%, showing an improvement of 1.01%. These results demonstrate that both ADSA and MCME algorithms are effective in enhancing the training performance of models.

On the F1-Score metric, the ADSA and MCME algorithms achieve the first and second positions in average results. The ADSA algorithm ranked first or second on 13 datasets, with an average result of 0.6868, which is an improvement of 0.19 over the original dataset's 0.6678. Notably, only the ADSA

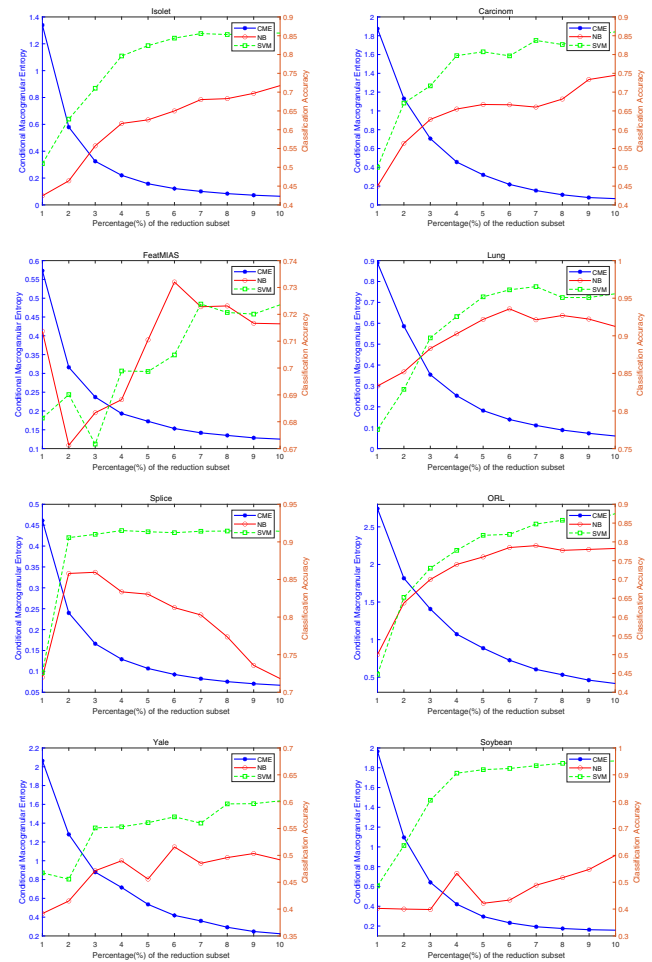


Fig. 7: Conditional Macrogranular Entropy and Percentage (10%) of the Feature Subset Accuracy Variation.

and MCME algorithms achieved average results better than the original dataset.

On the Recall metric, the ADSA algorithm ranks first in average results, while the MCME algorithm secures the second position. The ADSA algorithm achieved impressive results across 16 datasets, and notably, it is the only algorithm that surpassed the original data in terms of average results,

TABLE III: Classification accuracies of features subsets with NB (%)↑.

Datasets	ALL	NFRS	FSI	Gain-ratio	FSNCE	NSI	NFARNRS	MFIGI	FSFCE	MCME	ADSA
Carcinoma	68.98±15.00	72.9±11.92	17.24±7.36	21.74±12.42	59.26±6.76	60.27±10.69	64.89±9.31	60.41±11.87	70.06±7.25	74.47±10.05	76.32±11.99
Islet	86.35±2.74	62.05±3.98	80.26±2.58	71.41±3.68	72.31±1.47	56.03±3.55	62.76±1.95	58.21±4.12	72.05±3.20	71.73±2.97	77.69±4.09
Leukemia	45.48±5.55	77.86±15.68	45.48±5.55	45.48±5.55	84.52±10.42	77.86±15.68	77.86±15.68	87.62±12.28	51.43±9.06	86.19±12.28	87.62±11.49
ORL	81.75±4.87	34.75±4.63	40.50±5.24	73.00±6.21	72.25±8.93	72.00±5.99	74.75±5.71	59.50±10.72	71.25±5.03	78.25±5.78	75.75±6.24
Yale	48.09±7.82	45.46±8.72	41.25±12.50	39.8±9.06	43.50±11.29	43.45±12.27	50.90±9.11	44.57±6.83	49.80±12.68	49.17±5.59	52.65±11.06
FeatMIAS	69.49±9.20	65.49±8.41	70.11±6.18	69.8±9.22	70.10±7.99	71.07±8.94	71.7±10.68	72.69±6.94	69.79±7.08	71.65±7.84	73.79±10.95
Lung	85.14±9.00	91.18±7.15	81.21±9.02	86.4±8.08	86.79±8.36	90.75±5.56	90.23±7.42	95.16±6.39	93.70±5.03	91.25±7.43	91.68±7.58
Lung-Michigan	94.78±5.52	98.89±3.51	98.89±3.51	93.78±7.15	98.89±3.51	98.89±3.51	98.89±3.51	98.89±3.51	98.89±3.51	96.78±5.20	98.89±3.51
Ovarian	88.19±8.71	95.63±4.41	98.49±3.60	96.89±5.11	99.60±1.26	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	96.86±3.06
Seismic	41.25±7.90	93.42±0.01	91.64±0.97	79.33±4.60	91.45±1.01	93.42±0.01	93.42±0.01	93.42±0.01	91.49±1.15	93.42±0.01	93.42±0.01
Soybean	74.82±4.68	61.66±7.14	33.80±3.89	66.60±5.32	59.46±4.20	67.32±4.29	69.12±3.79	66.44±3.91	59.88±4.91	60.04±6.43	74.97±4.66
Solar-Flare	98.87±0.74	99.53±0.66	99.16±0.69	98.50±1.19	98.59±0.91	98.59±0.91	98.59±0.91	99.53±0.66	98.78±0.99	99.53±0.66	99.53±0.66
Pageblock	94.10±0.97	95.23±0.97	94.39±0.79	94.10±0.97	95.93±0.76	94.87±0.88	94.10±0.97	95.09±0.99	95.54±0.90	95.82±0.70	95.60±0.73
Splice	60.69±3.21	71.57±8.21	64.58±3.52	83.67±4.40	55.80±2.23	74.70±7.38	80.66±5.38	84.64±3.27	83.17±4.20	71.82±7.54	72.41±4.49
Nci9	32.14±16.99	43.81±21.12	18.45±12.00	47.14±17.16	17.86±9.61	38.33±23.87	47.50±16.09	36.07±13.19	53.69±20.49	50.83±23.26	68.21±12.69
Pixraw10P	77.00±14.94	99.00±4.83	99.00±3.16	99.00±3.16	97.00±4.83	96.00±6.99	97.00±4.83	92.00±9.19	99.00±3.16	94.00±6.99	93.00±6.75
Orlraws10P	58.00±13.98	94.00±8.43	72.00±7.89	89.00±8.76	86.00±11.74	89.00±11.01	80.00±10.54	81.00±8.76	90.00±9.43	87.00±11.60	92.00±10.33
GLI-85	80.42±11.59	84.31±11.77	90.42±11.08	83.61±14.80	88.47±12.85	94.03±8.20	93.06±7.97	92.78±6.24	96.39±5.83	94.31±7.98	85.69±11.03
Average	71.42±7.97	76.93±7.31	68.71±5.53	74.91±7.05	76.54±6.64	78.70±7.21	80.30±6.33	78.78±6.05	80.27±5.78	81.49±6.80	83.64±6.74

TABLE IV: Classification accuracies of features subsets with SVM (%)↑.

Datasets	ALL	NFRS	FSI	Gain-ratio	FSNCE	NSI	NFARNRS	MFIGI	FSFCE	MCME	ADSA
Carcinoma	91.30±7.47	86.62±6.99	28.17±6.23	79.02±9.74	60.77±10.86	70.57±8.48	71.61±10.76	66.17±6.16	74.61±8.59	86.03±9.41	86.55±8.54
Islet	95.83±1.58	79.49±4.72	93.53±2.86	82.18±4.25	79.04±3.01	65.51±4.13	75.83±3.52	45.77±4.87	79.62±4.09	85.71±2.47	85.90±3.34
Leukemia	87.62±9.44	88.81±11.06	95.48±7.31	91.9±8.56	86.19±11.49	88.81±11.06	88.81±11.06	86.19±10.24	90.95±10.34	92.38±9.93	89.52±10.52
ORL	95.00±3.54	31.25±3.39	40.75±4.72	81.5±5.92	80.5±8.06	73.5±5.43	71.75±5.01	36.00±6.03	74.50±7.43	87.50±4.71	87.25±5.71
Yale	74.73±7.08	40.05±13.65	37.60±10.19	49.6±11.87	48.07±12.26	46.9±16.04	46.13±14.79	33.99±13.68	45.42±10.46	60.15±8.59	56.66±10.55
FeatMIAS	70.87±9.86	69.22±5.67	67.96±7.87	70.16±9.41	71.96±6.17	68.9±3.84	71.48±7.23	64.32±4.85	73.20±7.04	72.35±5.55	74.16±6.87
Lung	94.16±5.93	93.54±5.60	89.80±5.08	89.19±6.39	85.29±6.91	91.73±3.77	92.7±4.67	84.31±4.92	94.70±4.81	95.59±5.72	95.54±4.88
Lung-Michigan	97.78±4.68	80±32.92	91.67±9.56	94.78±7.26	100.00±0.00	98.89±3.51	98.89±3.51	98.89±3.51	100.00±0.00	100.00±0.00	98.89±3.51
Ovarian	100.00±0.00	100.00±0.00	100.00±0.00	99.63±1.17	99.6±1.26	100.00±0.00	100.00±0.00	99.23±0.00	100.00±0.00	100.00±0.00	96.86±3.09
Seismic	66.15±31.29	93.42±0.01	49.19±34.66	61.75±34.33	72.6±30.53	93.42±0.01	93.42±0.01	93.42±0.01	56.45±36.35	93.42±0.01	93.42±0.01
Soybean	95.15±2.79	95.00±3.40	45.67±4.53	94.29±2.62	78.76±2.78	91.64±3.36	93.54±4.29	69.85±3.64	87.97±3.00	95.16±2.69	95.15±2.79
Solar-Flare	99.35±0.99	99.53±0.66	99.44±0.78	99.44±0.78	99.35±0.99	99.35±0.99	99.35±0.99	99.53±0.66	99.44±0.78	99.53±0.66	99.53±0.66
Pageblock	89.58±2.66	55.85±37.77	64.14±32.16	56.16±35.45	95.89±0.67	78.91±18.80	80.62±23.19	91.36±0.57	80.47±25.49	85.29±0.59	95.41±24.65
Splice	91.32±1.02	82.88±1.25	81.57±1.84	90.53±1.14	62.51±2.11	81.76±2.22	89.44±1.10	51.88±1.10	90.60±1.42	91.41±1.42	91.91±1.16
Nci9	57.26±20.93	47.86±19.78	28.21±12.35	75.48±16.50	22.26±11.31	36.43±24.97	55.83±31.10	42.86±14.13	55.95±27.50	53.57±21.45	68.21±19.68
Pixraw10P	99.00±3.16	99.00±3.16	99.00±3.16	98.00±6.32	93.00±10.59	98.00±6.32	98.00±4.22	86.00±6.75	96.00±5.16	98.00±4.22	99.00±3.16
Orlraws10P	98.00±4.22	92.00±11.35	64.00±11.74	92.00±6.32	93.00±10.59	84.00±8.43	82.00±10.33	63.00±15.06	86.00±9.66	92.00±9.19	94.00±9.66
GLI-85	30.52±4.83	60.00±16.70	68.33±11.87	45.83±20.14	75.42±13.94	86.81±12.22	79.86±15.18	65.69±38.62	60.00±13.49	66.67±34.17	80.56±27.55
Average	85.37±6.75	77.47±9.31	69.14±9.27	80.64±10.45	77.86±7.97	80.88±7.42	82.74±8.38	71.03±7.49	80.33±9.76	86.38±6.71	88.25±8.16

TABLE V: F1-Scores of feature subsets with algorithms ↑.

Datasets	ALL	NFRS	FSI	Gain-ratio	FSNCE	NSI	NFARNRS	MFIGI	FSFCE	MCME	ADSA
Carcinoma	0.7626	0.6998	0.1131	0.6584	0.4314	0.6323	0.5330	0.6664	0.6087	0.7225	0.7397
Islet	0.9555	0.7923	0.6627	0.8149	0.8124	0.7028	0.7509	0.7417	0.7872	0.8599	0.8589
Leukemia	0.4169	0.4604	0.4700	0.4507	0.4083	0.4604	0.4604	0.4590	0.4536	0.4490	0.4524
ORL	0.9408	0.2741	0.3083	0.7850	0.6169	0.6188	0.6743	0.4957	0.6940	0.8325	0.8206
Yale	0.6856	0.3149	0.2924	0.4607	0.3569	0.3836	0.4158	0.3736	0.3980	0.5178	0.5364
FeatMIAS	0.6963	0.6387	0.6896	0.7291	0.6941	0.7079	0.6975	0.7043	0.7048	0.7106	0.7450
Lung	0.8316	0.7919	0.6702	0.7607	0.6682	0.6797	0.7169	0.7943	0.7871	0.8442	0.8181
Lung-Michigan	0.4875	0.4857	0.4649	0.4822	0.4967	0.4967	0.4971	0.5000	0.5000	0.5000	0.4871
Ovarian	0.5000	0.4985	0.5000	0.4984	0.4985	0.5000	0.5000	0.5000	0.5000	0.5000	0.4880
Seismic	0.5279	0.5000	0.4854	0.5000	0.5262	0.5000	0.5000	0.0755	0.5000	0.5000	0.5000
Soybean	0.9598	0.9658	0.4213	0.9527	0.7595	0.8969	0.9305	0.9067	0.8502	0.9604	0.9598
Solar-Flare	0.3324	0.3326	0.3326	0.3324	0.3325	0.3325	0.3326	0.3325	0.3326	0.3326	0.3326
Pageblock	0.4853	0.4500	0.3594	0.4335	0.7570	0.3994	0.4746	0.7412	0.3325	0.6318	0.4403
Splice	0.9028	0.8150	0.7967	0.8925	0.4715	0.8157	0.8841	0.9015	0.4368	0.9055	0.9057
Nci9	0.3296	0.2500	0.1184	0.4926	0.1000	0.2148	0.3185	0.2685	0.8912	0.3319	0.4241
Pixraw10P	0.9867	0.9867	0.9867	0.9733	0.9200	0.9730	0.9733	0.9200	0.3556	0.9733	0.9767
Orlraws10P	0.9867	0.8867	0.5900	0.9083	0.8683	0.7983	0.7917	0.7950	0.9400	0.9150	0.9500
GLI-85	0.2323	0.6011	0.6806	0.3991	0.7020	0.7962	0.7023	0.5715	0.8283	0.6699	0.9169
Average	0.6678	0.5969	0.4968	0.6403	0.5775	0.6075	0.6196	0.6205	0.5947	0.6756	0.6868

TABLE VI: Recall of feature subsets with algorithms ↑.

Datasets	ALL	NFRS	FSI	Gain-ratio	F5CnCE	NSI	NFARNRS	MFIGI	FSFCE	MCME	ADSA
Carcinoma	0.7909	0.7318	0.1545	0.6864	0.4864	0.6652	0.5697	0.6879	0.6469	0.7470	0.7561
Islet	0.9558	0.7917	0.6641	0.8173	0.8154	0.7090	0.7513	0.7462	0.7891	0.8615	0.8603
Leukemia	0.4083	0.4667	0.4500	0.4833	0.4162	0.4667	0.4667	0.4667	0.4667	0.4500	0.4667
ORL	0.9550	0.3325	0.3750	0.8275	0.6825	0.6825	0.7250	0.5625	0.7425	0.8675	0.8575
Yale	0.7367	0.3767	0.3633	0.5200	0.4433	0.4467	0.4467	0.4433	0.4700	0.5933	0.6133
FeatMIAS	0.7053	0.6288	0.6938	0.7270	0.6964	0.6980	0.6921	0.6980	0.7029	0.7054	0.7517
Lung	0.8171	0.8047	0.6856	0.7580	0.6680	0.6710	0.7043	0.8004	0.7719	0.8442	0.8242
Lung-Michigan	0.4771	0.4813	0.4410	0.4764	0.4938	0.4938	0.4938	0.4944	0.5000	0.5000	0.4944
Ovarian	0.5000	0.5000	0.5000	0.4969	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000	0.5000
Seismic	0.4502	0.4830	0.3683	0.4830	0.4034	0.4830	0.4830	0.2059	0.4830	0.4830	0.4830
Soybean	0.9600	0.9664	0.4610	0.9519	0.7732	0.9032	0.9351	0.9075	0.8589	0.9595	0.9600
Solar-Flare	0.3327	0.3333	0.3333	0.3327	0.3330	0.3330	0.3330	0.3333	0.3330	0.3333	0.3333
Pagelbock	0.5484	0.5206	0.5142	0.5582	0.7233	0.5813	0.5536	0.6859	0.5426	0.6258	0.5446
Spice	0.9104	0.8392	0.8086	0.9074	0.5786	0.8544	0.9042	0.9167	0.9059	0.9185	0.9169
Nci9	0.3778	0.2889	0.1778	0.5111	0.1373	0.2556	0.3556	0.3111	0.3988	0.3667	0.4566
Pixraw10P	0.9900	0.9900	0.9900	0.9800	0.9400	0.9800	0.9800	0.9400	0.9500	0.9800	0.9800
Ortwrns10P	0.9900	0.9910	0.6400	0.9300	0.9000	0.8700	0.9300	0.9400	0.8700	0.9300	0.9600
GL-85	0.5000	0.6417	0.7083	0.5333	0.7400	0.8067	0.7300	0.5933	0.6733	0.7083	0.9250
Average	0.6892	0.6160	0.5183	0.6656	0.6003	0.6272	0.6403	0.6835	0.6288	0.6889	0.6403

due to an increased stopping parameter causing the selection process to terminate prematurely, leading to feature subsets that are too short to capture sufficient information. On the Orlraws10P dataset, there are slight fluctuations, but the overall performance remains robust. We conducted missing value experiments on five high-dimensional datasets. Due to space limitations, the result figures are included in the supplementary materials, where each color represents a different proportion of missing values. Overall, we observe that the feature subsets obtained by the ADSA

E. Statistical Analysis

We primarily used the Friedman Test [48] and the Nemenyi Post-Hoc Test [49] to further analyze the experimental results of our algorithms. These tests are validated based on the average ranking across different metrics.

We first calculate the average rankings of the algorithms for each evaluation metric, as shown in Tables III to VI. The results are presented in Table VII.

TABLE VII: The average rankings of each algorithm.

	SVM	NB	F1Score	Recall
NFRS	5.6944	5.9722	5.6111	5.6944
FSI	6.4444	7.3611	7.6667	8.0556
Gain-ratio	5.5000	7.1944	5.4722	4.8611
FSNCE	6.5278	6.5833	7.2500	6.8611
NSI	6.1389	5.7500	5.8611	6.3056
NFARNRS	5.7778	4.9722	5.5833	5.7778
MFIGI	8.0278	5.0000	5.4722	5.5833
FSFCE	5.1944	4.5833	5.9167	5.7778
MCME	<u>2.7778</u>	<u>4.1667</u>	<u>3.1389</u>	<u>3.0833</u>
ADSA	2.9167	3.4167	3.0278	3.0000

TABLE VIII: Values of F_F under four metrics.

	SVM	NB	F1Score	Recall	CV
F_F	6.5867	3.8646	5.3572	6.0788	1.6728

Firstly, the Friedman test [48] is used to determine whether there are overall significant differences in the performance rankings of multiple algorithms across multiple datasets. It assesses whether the overall performance differences are statistically significant. If significant differences are detected, further post-hoc tests such as the Nemenyi Post-Hoc Test are conducted. The null hypothesis is that there is no significant performance difference among all algorithms. The formula is defined as:

$$F_F = \frac{(N-1)\chi_F^2}{N(k-1)-\chi_F^2}, \quad (16)$$

where

$$\chi_F^2 = \frac{12N}{k(k+1)} \left(\sum_{j=1}^k R_j^2 - \frac{k(k+1)^2}{4} \right), \quad (17)$$

where k is the number of algorithms and N is the number of datasets. R_j represents the average ranking value of j -th algorithm on different metric. The calculated significance value $CV = 1.6728$. The performance differences between our algorithm and others are evaluated using the Friedman Test

(F_F). With $\alpha = 0.1$, the metric values in Table VIII exceed the critical value ($CV = 1.6728$), indicating significant performance differences among ten feature selection algorithms.

Nemenyi Post-Hoc Test [49]: Based on a significant Friedman test result, the Nemenyi Post-Hoc Test is used to identify which specific algorithm pairs exhibit significant differences. It calculates a Critical Difference (CD), and if the average rank difference between two algorithms exceeds this CD, the difference is considered statistically significant. In the context of feature selection, the differences among algorithms can be visually illustrated using a Critical Difference Diagram as Fig 8. If the average rank difference between two algorithms in Table VII exceeds the critical difference (CD), the hypothesis of equal performance between the algorithms is rejected at the given confidence level $\alpha = 0.1$. Critical difference CD values are calculated as follows:

$$CD = q_\alpha \sqrt{\frac{k(k+1)}{6N}}, \quad (18)$$

where q_α is a critical value [49] by checking the table we can get that in this paper $q_\alpha = 2.920$ when $\alpha = 0.1$. k is the number of algorithms and N is the number of datasets. For a confidence level of $\alpha = 0.1$, the critical difference CD is calculated as 2.9468.

In the results of the Nemenyi post-hoc test, ADSA algorithm ranks first across all metrics, MCME algorithm ranks second as shown in Fig 8. The results indicate that both ADSA and MCME algorithms exhibit significant differences compared to the other algorithms and outperform them clearly.

Based on the results of the Friedman Test and Nemenyi Post-Hoc Test, we verify that our algorithms have significant differences compared to other algorithms, thereby validating the rationality of the macrogranular entropy measurement and further demonstrate the advantage of the degree of symmetric association in measuring feature significance.

F. Discussion

In this subsection, we provide a more comprehensive discussion of the ADSA algorithm, analyzing the advantages and limitations of the symmetric association degree from multiple perspectives, which can serve as guidance for future work. Based on the experimental performance and the results of statistical tests, the ADSA algorithm demonstrates excellent overall performance and performs well on datasets from various domains such as image and biology. However, one issue lies in the fact that the degree of symmetric association evaluates the significance of a feature by considering its independence within the candidate feature space, its association with the selected feature space, and its relevance to the decision attribute. As a result, the algorithm exhibits relatively high computational complexity $O(n^2 \times ((2M - k) \times k/2))$, which may be computationally expensive when dealing with high-dimensional datasets. Future work could focus on improving computational efficiency, for instance, by pre-grouping features or eliminating a portion of redundant features before applying the ADSA algorithm for feature selection.

Additionally, we have conducted experiments on the accuracy of feature subsets selected by the ADSA algorithm



Fig. 8: The results of Nemenyi post-hoc test.

and macro-granular entropy across different classifiers and under varying levels of noise in high-dimensional datasets. The results are presented in bar charts. Due to space constraints, these figures are included in the supplementary materials. The figures indicate that the ADSA algorithm exhibits noticeable fluctuations on some datasets, suggesting that future work may explore preprocessing techniques for noise reduction to enhance the stability of the algorithm.

Gene expression data is often characterized by thousands of features (genes) with only a small number of samples. Our method is particularly suitable for such high-dimensional and complex data. ADSA's ability to reduce dimensionality while preserving relevant information helps researchers pinpoint critical genes associated with specific diseases, aiding in drug development and personalized medicine.

V. CONCLUSIONS AND FUTURE WORK

In practical feature selection tasks, constructing an effective measurement to quantify the relationship between the information content of features is essential. Fuzzy information entropy and its generalized forms are commonly used in feature selection, but they have limitations in measuring the relationship between features and the decision, particularly the lack of monotonicity in fuzzy conditional entropy. The monotonicity of conditional entropy is crucial, as it reflects changes in decision uncertainty during the feature selection process. To address the challenges of fuzzy information entropy, we introduce a new measurement Macrogranular Entropy and some generalized forms, the conditional macrogranular entropy is monotonic. Macrogranular entropy possesses favorable properties that fuzzy information entropy lacks, making it more effective in measuring feature information in real-valued feature selection tasks. Not only fuzzy information entropy, but also other measurements such as combination entropy, self-information, measure the relationship between different features by performing intersection operations on the fuzzy relations formed by features. In contrast, macrogranular entropy measurement adopts a union operation on fuzzy relations, considering a broader scope.

Macrogranular entropy effectively quantifies feature information even without explicit sample relationships, making it ideal for feature selection in incomplete or semi-supervised situations with limited data. This could be followed by further research in this area. This is especially valuable in fields like medical diagnostics, where acquiring comprehensive data is costly and sample relationships may be unclear. Although the ADSA algorithm demonstrates good performance in general, there is still some sensitivity to noise in certain datasets.

Future studies can explore integrating noise filtering or robust estimation techniques to improve the stability of feature selection under noisy conditions. ADSA algorithm involves evaluating symmetric association among features, the current time complexity is relatively high when applied to extremely high-dimensional datasets. Future work may focus on reducing computational cost through strategies such as feature pre-grouping, incremental selection. Real-world data is often dynamic and continuously generated. Therefore, adapting ADSA to handle online learning or streaming feature selection scenarios will be an important direction for enhancing its practical utility.

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